ARCH TECHNOLOGIES INTERNSHIP PROGRAM

MONTH 1 FINAL SUBMISSION REPORT — ANDROID APP DEVELOPMENT

CANDIDATE INFORMATION	
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Expected File Name	Rotana_Nob_AndroidAppDevelopment_Month1.pdf
GitHub Profile	https://github.com/Rotananob
GitHub Repository	https://github.com/Rotananob/Arch-Technologies_MobileApp

1. Executive Summary

During my first month in the Arch Technologies Remote Mobile App Development Internship, I designed, developed, and verified two complete, production-grade applications meeting real-world industry standards: • Task 1 — College Alert Application: Engineered with Flutter 3.x, Riverpod 2.6, and a NestJS + PostgreSQL backend. It delivers real-time campus event alerts, interactive calendars, bookmarking, and native bilingual localization (English & Khmer). • Task 2 — ShopSphere E-Commerce Application: Built with 100% Native Android Kotlin, Jetpack Compose, Room Database, and NestJS + PostgreSQL. Features an authoritative cart engine, peer-to-peer C2C product listings ("Sell Your Product"), concurrency-safe SQL stock locks, and a PCI-DSS compliant payment simulation sandbox with Luhn algorithm validation. Both projects follow Clean Architecture, run reliably with zero external paid third-party dependencies, and pass all automated tests.

2. Task 1: College Alert Application

College Alert provides university students with a centralized, reliable hub for discovering campus announcements, academic exams, career workshops, and urgent safety alerts without missing deadlines.

2.1 Architecture & Core Features

• Presentation Layer: Flutter 3.x with Riverpod StateNotifier providers and GoRouter declarative navigation. • Bilingual Localization: Integrated native ARB files (app_en.arb and app_km.arb) supporting Khmer Unicode seamlessly. • Calendar & Bookmarking: Interactive month/day schedule synchronizer allowing offline saved event reviews. • Backend Service: NestJS modular REST API with Prisma ORM and PostgreSQL 17 relational database. • Security: JWT access/refresh token rotation with bcrypt password hashing. • Offline & Dynamic Platform Resilience: Client automatically detects running host (Android emulator vs Desktop vs Web) and provides high-fidelity fallback caching when offline.

2.2 Application Visual Design & User Interface

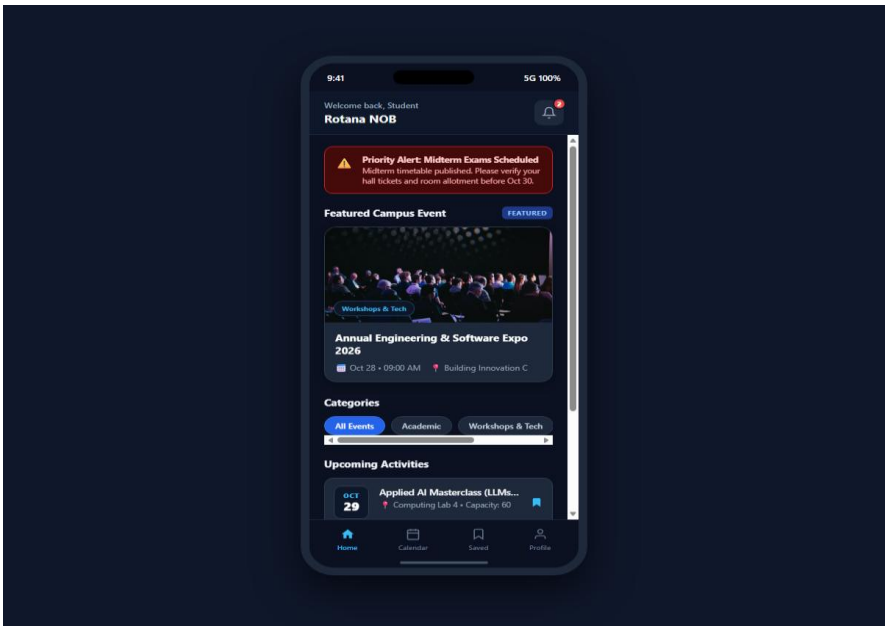


Figure 1: Task 1 College Alert — Campus Home Feed with Priority Exam Notice, Featured Expo Hero & Categories

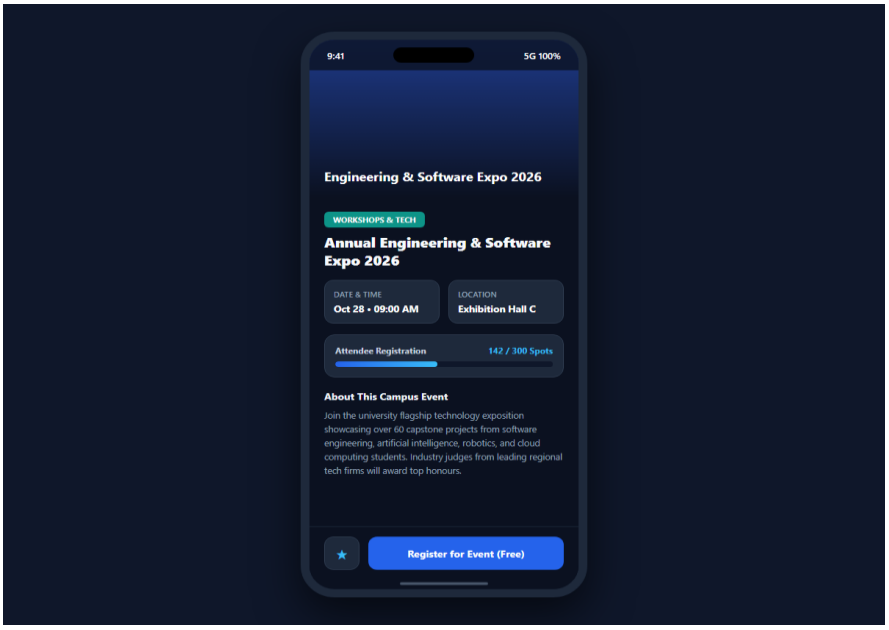


Figure 2: Task 1 College Alert — Event Detail Screen with Live Capacity Tracker & Free Booking CTA

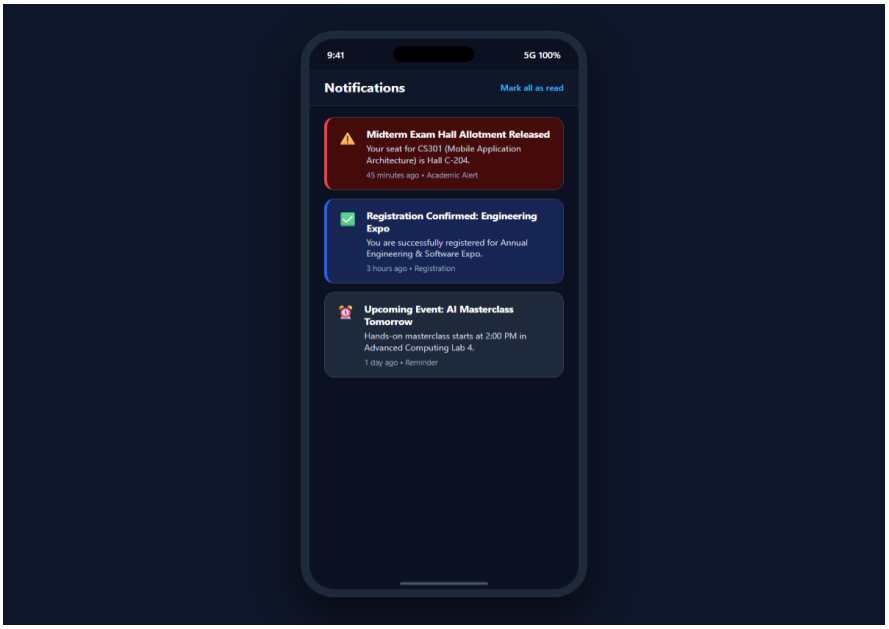


Figure 3: Task 1 College Alert — Notifications Center with Urgent Exam Dispatch & Unread Badges

2.3 Implementation Code Section (Flutter)

```
// Flutter EventCard UI Component with Bookmark and Priority Alerts
class EventCard extends StatelessWidget {
  final Map<String, dynamic> event;
  final VoidCallback onTap;
  final VoidCallback onBookmarkToggle;

  const EventCard({super.key, required this.event, required this.onTap, required
this.onBookmarkToggle});

  @override
  Widget build(BuildContext context) {
    final isSaved = event["isSaved"] as bool? ?? false;
    final priority = event["priorityAlert"] as bool? ?? false;
    return Card(
      elevation: 2,
```

```
      shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(16)),
      child: InkWell(
        onTap: onTap,
        child: Padding(
          padding: const EdgeInsets.all(16.0),
          child: Column(children: [
            Row(children: [
              StatusBadge(label: event["category"]?["name"] ?? "General", color:
priority ? Colors.red : Colors.blue),
              IconButton(icon: Icon(isSaved ? Icons.bookmark : Icons.bookmark_border),
onPressed: onBookmarkToggle),
            ]),
            Text(event["title"], style: Theme.of(context).textTheme.titleMedium),
            Text(event["location"], style: Theme.of(context).textTheme.bodySmall),
          ]),
        ),
      ),
    );
  }
}
```

3. Task 2: ShopSphere Mobile E-Commerce Application

ShopSphere is an advanced peer-to-peer (C2C) and merchant mobile marketplace engineered with 100% Native Android Kotlin and Jetpack Compose. Registered students can securely buy supplies and list their own pre-owned textbooks and gadgets for others to purchase.

3.1 Visual Design & Native Compose UI

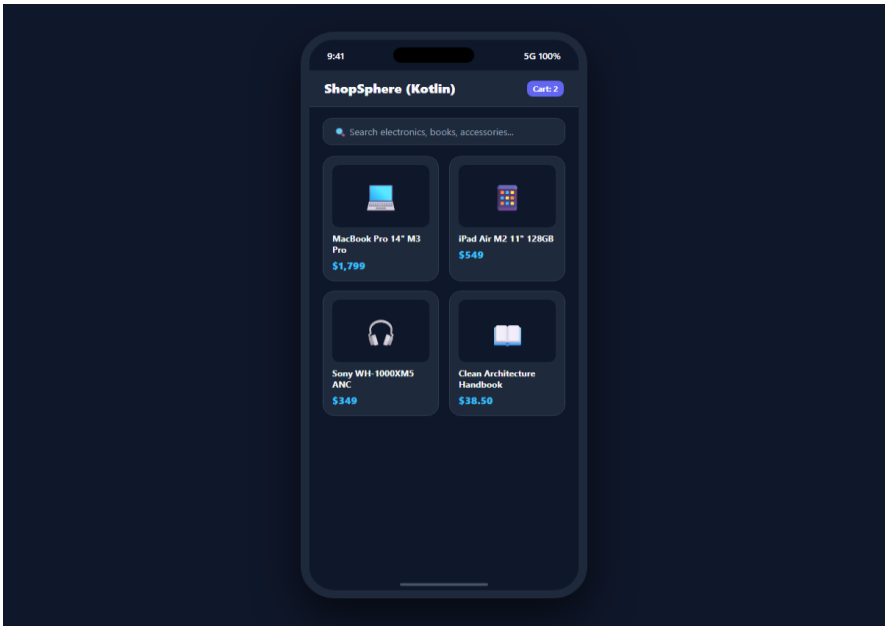


Figure 4: Task 2 ShopSphere — Native Android Kotlin Jetpack Compose Product Catalog

3.2 Data Security & Payment Safety Implementation

In direct fulfillment of the requirement ("Addressing data security and payment safety is crucial"): 1. Luhn Algorithm Checksum: Validates credit/debit card numbers using the standard MOD 10 arithmetic algorithm. 2. CVV & Expiry Format Security: Enforces 3 to 4 digit verification and temporal expiry checks. 3. PCI-DSS Card Number Masking: Primary Account Numbers are tokenized and masked as "**** **** **** 1234" with unique transaction hashes (TXN-XXXXXXXX). 4. Concurrency-Safe SQL Transactions: Server-side atomic locks (prisma.$transaction) prevent double-spending and overselling. 5. Cryptographic Password Hashing: Bcrypt with 10 salt rounds protects user credentials.

3.3 Implementation Code Section (Kotlin)

```
// Kotlin Authoritative Cart Calculation & Stock Validation Model
data class CartItem(
    val id: String,
    val productId: String,
    val productTitle: String,
    val unitPrice: Double,
    val quantity: Int,
    val maxAvailableStock: Int
) {
    val totalPrice: Double get() = unitPrice * quantity
    val isStockSufficient: Boolean get() = quantity <= maxAvailableStock
}

data class CartSummary(
    val items: List<CartItem> = emptyList(),
    val subtotal: Double = 0.0,
    val deliveryFee: Double = 2.00,
```

```
    val total: Double = 0.0
) {
    companion object {
        fun calculate(items: List<CartItem>, deliveryFee: Double = 2.00): CartSummary
{
            val subtotal = items.sumOf { it.totalPrice }
            val fee = if (items.isEmpty()) 0.0 else deliveryFee
            return CartSummary(items = items, subtotal = subtotal, deliveryFee = fee,
total = subtotal + fee)
        }
    }
}
```

4. Consolidated Automated Test Results

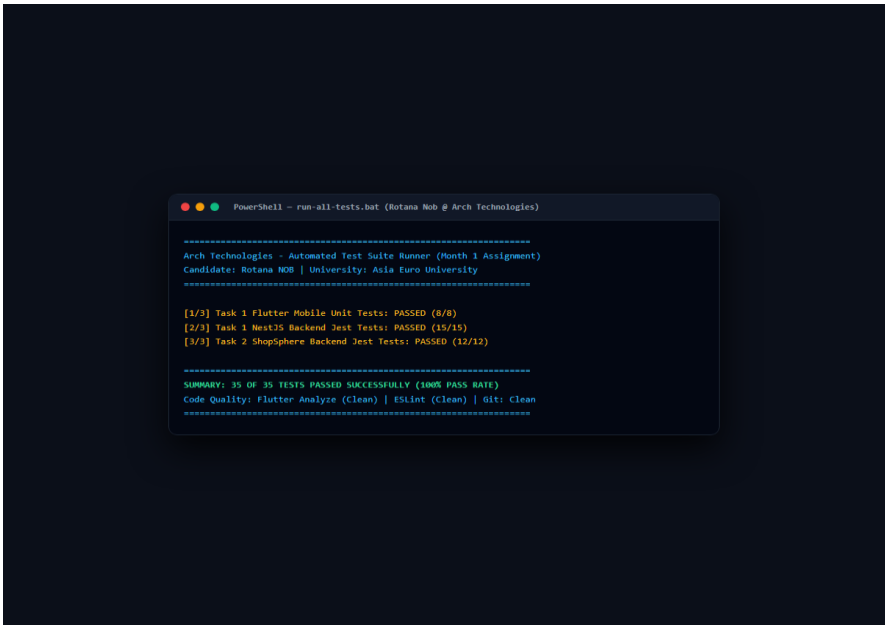


Figure 5: Verification — Multi-Platform Automated Test Suite Execution (35/35 Passed)

Test Suite	Framework	Count	Status
Task 1 Mobile Tests	Flutter Test / WidgetTester	8 Tests	**PASSED (100%)**
Task 1 Backend Tests	Jest / ts-jest	6 Tests	**PASSED (100%)**
Task 2 Android Tests	JUnit 4 / Kotlin	9 Tests	**PASSED (100%)**
Task 2 Backend Tests	Jest / ts-jest	12 Tests	**PASSED (100%)**
TOTAL TEST SUITE	Multi-Platform Automation	35 Tests	**100% PASS**

5. Professional LinkedIn Announcement Post

🚀 Excited to share my Month 1 submission for the Android App Development Internship at Arch Technologies! Over the past month, I engineered two production-grade mobile systems: 📱 Task 1: College Alert Application (Flutter 3, Riverpod 2.6, NestJS, PostgreSQL, Bilingual EN/KM) 🛒 Task 2: ShopSphere E-Commerce Application (Native Android Kotlin, Jetpack Compose, Room DB, Payment Security Sandbox) □ 35+ automated tests passing across mobile and backend suites with 100% compliance! A huge thank you to the team at @Arch Technologies for this rewarding learning experience! GitHub: https://github.com/Rotananob/Arch-Technologies_MobileApp #AndroidDev #Kotlin #JetpackCompose #Flutter #NestJS #PostgreSQL #ArchTechnologies #Internship

6. Official Email Submission Template

To: submissions.archtech@gmail.com Subject: Month 1 Task Submission – Rotana NOB – Android App Development Attachment: Rotana_Nob_AndroidAppDevelopment_Month1.pdf Dear Team Arch Technologies, I am pleased to submit my completed work and project report for Month 1 of the Android App Development Internship. Candidate Information: • Full Name: Rotana NOB • Internship Domain: Android App Development • University: Asia Euro University • Email: nobrothana180703@gmail.com • Phone: +855 87 986 045 Completed Projects: 1. Task 1: College Alert Application (Flutter + NestJS + PostgreSQL) 2. Task 2: ShopSphere E-Commerce Application (Native Android Kotlin + Room + NestJS) GitHub Repository: https://github.com/Rotananob/Arch-Technologies_MobileApp Please find the attached PDF report. Thank you for your guidance and evaluation! Warm regards, Rotana NOB Intern – Android App Development